Kafka

1_Разворачиваем кластер виртуальных машин с помощью файла docker-compose_cluster.yml

```
[yc-user@kafka-node:~$ sudo docker ps -a
CONTAINER ID   IMAGE                               COMMAND                  CREATED          STATUS          PORTS
                                                                               NAMES
24d7e4132556   clickhouse/clickhouse-server        "/entrypoint.sh"         25 seconds ago   Up 23 seconds   9009/tcp, 0.0.0.0:9124->8123/tcp, [::]:9124->8123/tcp, 0.0.0.0:8003
->9000/tcp, [::]:8003->9000/tcp, 0.0.0.0:9365->9363/tcp, [::]:9365->9363/tcp   ch2
1e40b863bb01   clickhouse/clickhouse-server        "/entrypoint.sh"         25 seconds ago   Up 23 seconds   9009/tcp, 0.0.0.0:9123->8123/tcp, [::]:9123->8123/tcp, 0.0.0.0:8002
->9000/tcp, [::]:8002->9000/tcp, 0.0.0.0:9364->9363/tcp, [::]:9364->9363/tcp   ch1
c0f0b0c9b540   clickhouse/clickhouse-server        "/entrypoint.sh"         25 seconds ago   Up 23 seconds   9009/tcp, 0.0.0.0:9125->8123/tcp, [::]:9125->8123/tcp, 0.0.0.0:8004
->9000/tcp, [::]:8004->9000/tcp, 0.0.0.0:9366->9363/tcp, [::]:9366->9363/tcp   ch3
97364acd59ff   confluentinc/cp-kafka:7.8.0         "/etc/confluent/dock…"   25 seconds ago   Up 23 seconds   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:9999->9999/tcp
, [::]:9999->9999/tcp, 0.0.0.0:29092->29092/tcp, [::]:29092->29092/tcp         kafka
d2d812ead29f   clickhouse/clickhouse-server        "/entrypoint.sh"         25 seconds ago   Up 23 seconds   9009/tcp, 0.0.0.0:9126->8123/tcp, [::]:9126->8123/tcp, 0.0.0.0:8005
->9000/tcp, [::]:8005->9000/tcp, 0.0.0.0:9367->9363/tcp, [::]:9367->9363/tcp   ch4
c538a003e54a   grafana/grafana-enterprise:latest   "/run.sh"                25 seconds ago   Up 23 seconds   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
                                                                               grafana
c063a0305560   zookeeper:latest                    "/docker-entrypoint.…"   25 seconds ago   Up 23 seconds   0.0.0.0:2888->2888/tcp, [::]:2888->2888/tcp, 0.0.0.0:3888->3888/tcp
, [::]:3888->3888/tcp, 8080/tcp, 0.0.0.0:2182->2181/tcp, [::]:2182->2181/tcp   zk
8dc33e684b10   prom/prometheus:latest              "/bin/prometheus --c…"   25 seconds ago   Up 23 seconds   0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
                                                                               prom
```

За Kafka отвечает этот кусок:

```yaml
# ================================================
# Kafka
# ================================================
kafka:
  image: confluentinc/cp-kafka:7.8.0
  hostname: kafka
  container_name: kafka
  ports:
    - "9092:9092"
    - "29092:29092"
    - "9999:9999"
  environment:
    KAFKA_ADVERTISED_LISTENERS: INTERNAL://kafka:19092,EXTERNAL://${DOCKER_HOST_IP:-127.0.0.1}:9092,DOCKER://host.docker.internal:29092
    KAFKA_LISTENER_SECURITY_PROTOCOL_MAP: INTERNAL:PLAINTEXT,EXTERNAL:PLAINTEXT,DOCKER:PLAINTEXT
    KAFKA_INTER_BROKER_LISTENER_NAME: INTERNAL
    KAFKA_ZOOKEEPER_CONNECT: "zookeeper:2181"
    KAFKA_BROKER_ID: 1
    KAFKA_LOG4J_LOGGERS: "kafka.controller=INFO,kafka.producer.async.DefaultEventHandler=INFO,state.change.logger=INFO"
    KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
    KAFKA_TRANSACTION_STATE_LOG_REPLICATION_FACTOR: 1
    KAFKA_TRANSACTION_STATE_LOG_MIN_ISR: 1
    KAFKA_JMX_PORT: 9999
    KAFKA_JMX_HOSTNAME: ${DOCKER_HOST_IP:-127.0.0.1}
    KAFKA_AUTHORIZER_CLASS_NAME: kafka.security.authorizer.AclAuthorizer
    KAFKA_ALLOW_EVERYONE_IF_NO_ACL_FOUND: "true"
  networks:
    - otus_ch_net
  depends_on:
    - zookeeper
```

Создаем топик

```
docker exec -it kafka bash
[appuser@kafka ~]$ kafka-topics --bootstrap-server kafka:9092 --create --topic task_17 --partitions 2 --replication-factor 1
WARNING: Due to limitations in metric names, topics with a period ('.') or underscore ('_') could collide. To avoid issues it is best to use either, but not both.
Created topic task_17.
```

Вставляем туда данные

```
for i in {1..20}; do
```

```
>   echo "{\"id\":$i, \"data\":\"message_$i\", \"load_ts\":\"$(date -u +'%Y-%m-%d %H:%M:%S')\"}"
> done > messages.json
[appuser@kafka ~]$
[appuser@kafka ~]$ kafka-console-producer --bootstrap-server kafka:9092 --topic task_17 <
messages.json
```

Проверяем их наличие

```
kafka-console-consumer --bootstrap-server kafka:9092 --topic task_17 --from-beginning
{"id":1, "data":"message_1", "load_ts":"2025-12-04 15:01:16"}
{"id":2, "data":"message_2", "load_ts":"2025-12-04 15:01:16"}
{"id":3, "data":"message_3", "load_ts":"2025-12-04 15:01:16"}
{"id":4, "data":"message_4", "load_ts":"2025-12-04 15:01:16"}
{"id":5, "data":"message_5", "load_ts":"2025-12-04 15:01:16"}
{"id":6, "data":"message_6", "load_ts":"2025-12-04 15:01:16"}
{"id":7, "data":"message_7", "load_ts":"2025-12-04 15:01:16"}
{"id":8, "data":"message_8", "load_ts":"2025-12-04 15:01:16"}
{"id":9, "data":"message_9", "load_ts":"2025-12-04 15:01:16"}
{"id":10, "data":"message_10", "load_ts":"2025-12-04 15:01:16"}
{"id":11, "data":"message_11", "load_ts":"2025-12-04 15:01:16"}
{"id":12, "data":"message_12", "load_ts":"2025-12-04 15:01:16"}
{"id":13, "data":"message_13", "load_ts":"2025-12-04 15:01:16"}
{"id":14, "data":"message_14", "load_ts":"2025-12-04 15:01:16"}
{"id":15, "data":"message_15", "load_ts":"2025-12-04 15:01:16"}
{"id":16, "data":"message_16", "load_ts":"2025-12-04 15:01:16"}
{"id":17, "data":"message_17", "load_ts":"2025-12-04 15:01:16"}
{"id":18, "data":"message_18", "load_ts":"2025-12-04 15:01:16"}
{"id":19, "data":"message_19", "load_ts":"2025-12-04 15:01:16"}
{"id":20, "data":"message_20", "load_ts":"2025-12-04 15:01:16"}
```

Создаем на стороне ClickHouse таблицу приемник и материализованное представление

```sql
CREATE DATABASE IF NOT EXISTS task_17;

CREATE TABLE IF NOT EXISTS task_17.kafka
(
    id UInt32,
    data StringCREATE DATABASE IF NOT EXISTS task_17;

CREATE TABLE IF NOT EXISTS task_17.kafka
(
    id UInt32,
    data String,
    load_ts DateTime
)
    ENGINE = Kafka
        SETTINGS
            kafka_broker_list = 'kafka:19092',
            kafka_topic_list = 'task_17',
            kafka_group_name = 'ch_group_3',
            kafka_format = 'JSONEachRow';

CREATE TABLE IF NOT EXISTS task_17.data
(
    id      UInt32,
    data    String,
    load_ts DateTime
)
    ENGINE = MergeTree
        ORDER BY (load_ts, id);

CREATE MATERIALIZED VIEW task_17.data_mv TO task_17.data AS
SELECT id,
       data,
       load_ts
  FROM task_17.kafka;

CREATE DATABASE IF NOT EXISTS task_17

Query id: afd3b827-4a00-49c0-8c16-f42476842eec
```

```
Ok.

0 rows in set. Elapsed: 0.008 sec.


CREATE TABLE IF NOT EXISTS task_17.kafka
(
    `id` UInt32,
    `data` String,
    `load_ts` DateTime
)
ENGINE = Kafka
SETTINGS kafka_broker_list = 'kafka:19092', kafka_topic_list = 'task_17', kafka_group_name =
'ch_group_3', kafka_format = 'JSONEachRow'

Query id: 94e81bda-0727-4bcb-baef-346b05ebaf56

Ok.

0 rows in set. Elapsed: 0.006 sec.


CREATE TABLE IF NOT EXISTS task_17.data
(
    `id` UInt32,
    `data` String,
    `load_ts` DateTime
)
ENGINE = MergeTree
ORDER BY (load_ts, id)

Query id: 01529d97-fba4-44b0-a8ce-b06ad1ab7a63

Ok.

0 rows in set. Elapsed: 0.011 sec.


CREATE MATERIALIZED VIEW task_17.data_mv TO task_17.data
AS SELECT
    id,
    data,
    load_ts
FROM task_17.kafka

Query id: a4584bb6-3521-4d16-8566-a6e988a94ab8

Ok.

0 rows in set. Elapsed: 0.010 sec.
```

Проверяем чтение данных:

```
SELECT *
  FROM task_17.data;

SELECT *
FROM task_17.data

Query id: 58b41725-324f-4d68-897a-34f207321b02

   ┌─id─┬─data──────┬─────────────load_ts─┐
1. │  1 │ message_1 │ 2025-12-04 15:01:16 │
2. │  2 │ message_2 │ 2025-12-04 15:01:16 │
3. │  3 │ message_3 │ 2025-12-04 15:01:16 │
4. │  4 │ message_4 │ 2025-12-04 15:01:16 │
5. │  5 │ message_5 │ 2025-12-04 15:01:16 │
6. │  6 │ message_6 │ 2025-12-04 15:01:16 │
7. │  7 │ message_7 │ 2025-12-04 15:01:16 │
```

```
 8. │  8 │ message_8   │ 2025-12-04 15:01:16 │
 9. │  9 │ message_9   │ 2025-12-04 15:01:16 │
```

Также проверяем данные в системной таблице

```
SELECT *
  FROM system.kafka_consumers
FORMAT Vertical;

SELECT *
FROM system.kafka_consumers
FORMAT Vertical

Query id: 0f845fd5-53a8-4698-ab40-cc5e0fe910a8

Row 1:
──────
database:                    task_17
table:                       kafka
consumer_id:                 ClickHouse-clickhouse1-task_17-kafka-a5d0aa54-6595-4da0-bf90-
ad51d794be8a
assignments.topic:           ['task_17','task_17']
assignments.partition_id:    [0,1]
assignments.current_offset: [20,-1001]
assignments.intent_size:     [NULL,NULL]
exceptions.time:             []
exceptions.text:             []
last_poll_time:              2025-12-04 18:06:54
num_messages_read:           20
last_commit_time:            2025-12-04 18:04:42
num_commits:                 1
last_rebalance_time:         1970-01-01 03:00:00
num_rebalance_revocations:   0
num_rebalance_assignments:   1
is_currently_used:           1
last_used:                   2025-12-04 18:06:49.620273
rdkafka_stat:

1 row in set. Elapsed: 0.003 sec.
```

Добавляем еще одно сообщение

```
kafka-console-producer --bootstrap-server kafka:9092 --topic task_17
>{"id":20, "data":"NEW_message", "load_ts":"2025-12-04 21:00:00"}
```

И видем его в clickhouse

```
SELECT *
  FROM task_17.data;

SELECT *
FROM task_17.data

Query id: 9766abbe-61da-43e3-9855-6867ad19aabd

    ┌─id─┬─data─────────┬─────────────load_ts─┐
 1. │  1 │ message_1    │ 2025-12-04 15:01:16 │
 2. │  2 │ message_2    │ 2025-12-04 15:01:16 │
 3. │  3 │ message_3    │ 2025-12-04 15:01:16 │
 4. │  4 │ message_4    │ 2025-12-04 15:01:16 │
 5. │  5 │ message_5    │ 2025-12-04 15:01:16 │
 6. │  6 │ message_6    │ 2025-12-04 15:01:16 │
 7. │  7 │ message_7    │ 2025-12-04 15:01:16 │
 8. │  8 │ message_8    │ 2025-12-04 15:01:16 │
 9. │  9 │ message_9    │ 2025-12-04 15:01:16 │
10. │ 10 │ message_10   │ 2025-12-04 15:01:16 │
```

11	message_11	2025-12-04 15:01:16
12	message_12	2025-12-04 15:01:16
13	message_13	2025-12-04 15:01:16
14	message_14	2025-12-04 15:01:16
15	message_15	2025-12-04 15:01:16
16	message_16	2025-12-04 15:01:16
17	message_17	2025-12-04 15:01:16
18	message_18	2025-12-04 15:01:16
19	message_19	2025-12-04 15:01:16
20	message_20	2025-12-04 15:01:16
20	NEW_message	2025-12-04 21:00:00